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BPSC SPECIAL NOTES

TRE 4.0 Computer Science

Operating Systems (Part 1/4)

OS: UGC NET और TRE में सबसे ज्यादा पूछा जाने वाला topic

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OS Part 1/4 — Intro

1 Operating System (OS) क्या है

- Operating System = वह system software जो computer hardware और user/application programs के बीच interface का काम करता है.
  - → Hardware (CPU, memory, I/O) को manage करता है और बाकी software को चलने का platform देता है.
  - → उदाहरण: Windows, Linux, macOS, Android, UNIX. (Oracle OS नहीं, DBMS (Database Management System) है.)
- Exam Point: "Software जो hardware पर चलकर दूसरे software के लिए platform बने" = Operating System (TRE 2.0 Q83); System software बड़ा समूह है (utilities, translators भी).

2 OS के मुख्य functions (NCERT XI Ch1)

- Process management — process बनाना, schedule करना, terminate करना.
- Memory management — RAM (Random Access Memory) allocate/free करना, virtual memory.
- File management — files/directories बनाना, delete, rename, protect.
- Device (I/O — Input/Output) management — drivers, buffering, spooling.
- Security & protection — user authentication, access control.
- User interface — CLI (Command Line Interface) और GUI (Graphical User Interface).

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Types of OS

1 Batch OS

- Users computer से सीधे interact नहीं करते; jobs का batch operator चलाता है.
  - → फायदा: बड़े repetitive jobs में अच्छा CPU use. नुकसान: कोई interaction नहीं, CPU idle रह सकता है, debugging कठिन.

2 Multiprogramming OS

- Multiprogramming = कई programs एक साथ main memory में; CPU एक program के I/O wait पर तुरंत दूसरे को दे दी जाती है.
  - → लक्षण: (a) एक से ज़्यादा program memory में, (b) event का wait होते ही दूसरा program schedule, (c) terminate होते ही दूसरा schedule — तीनों सही (NET Dec 2012 P3).

3 Time-sharing (Multitasking) OS

- CPU time छोटे time slices/quantum में सब users/processes में बँटता है (Round Robin); response time कम.
  - → हर user को लगता है कि पूरा computer उसी का है. Example: UNIX, Linux.

4 Multiprocessor / Parallel, Distributed, Network, Real-time, Embedded

- Multiprocessor: एक से ज़्यादा CPU, shared memory; throughput और reliability ज़्यादा.
- Distributed OS: कई अलग computers एक single system जैसे दिखते हैं; resources share होते हैं.

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OS Structure & Kernel

1 Kernel, Shell और Modes

- Kernel = OS का core; system call interface के नीचे और hardware के ऊपर का सारा हिस्सा.
- Shell = command interpreter, user और kernel के बीच (UNIX: Bourne sh, C shell csh, Korn ksh, Bash).
- Dual mode: User mode (limited) और Kernel/Supervisor mode (privileged instructions). Mode bit: 1 = user, 0 = kernel.
  - → Signal handler, system call, interrupt आने पर CPU kernel mode में जाती है.
  - → User-kernel mode switch का समय < दो processes के बीच context switch का समय (T1 < T2).

2 OS structures

- Monolithic: पूरा OS एक बड़ा kernel (पुराना UNIX, Linux) — तेज़ पर bug-prone.
- Layered: layer 0 hardware ... top = user interface; debugging आसान, पर performance थोड़ी कम.
- Microkernel: सिर्फ ज़रूरी चीज़ें (IPC — Inter-Process Communication, basic scheduling, memory) kernel में; बाकी user space में (Mach, QNX).
- Modular/Hybrid: Windows NT, macOS. Windows NT को object-oriented कहा जाता है (kernel objects सिर्फ methods से).
- Virtual machine: एक hardware पर कई OS (जैसे VMware).

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UNIX & Windows Basics

1 UNIX commands (NET में बार-बार)

- ls (list), cat (file दिखाना), cp (copy), mv (move/rename), rm (delete), mkdir, cd, pwd, grep, sort, head, tail, wc.
- nice = process की priority set; kill = signal भेजना; ps = processes list; init = पहला process (PID 1).
- Redirection: < input file से, > output file में, | pipe एक का output दूसरे का input.
  - → sort <in >temp; head -30 <temp; rm temp - in को sort करके temp में लिखो, temp की 30 lines print करो, फिर temp हटाओ.

2 UNIX file system एक झलक

- inode में file metadata + 10 direct, 1 single, 1 double, 1 triple indirect pointers.
- Max file size = (10 + n + n^2 + n^3) x block size, n = block size / address size.
  - → Worked example: block 1 KB, address 4 B to n = 256; (10+256+65536+16777216) KB ≈ 16 GB.

3 Windows basics

- Windows XP/Vista file systems: FAT-16 (File Allocation Table), FAT-32, NTFS (New Technology File System).
- Hibernate: RAM की state disk पर save, power off, programs खुले रहते हैं; Sleep: low power RAM में. Shutdown: सब बंद.
- Windows NT का kernel object-based है.

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Numericals — Detailed Solutions

(1) NET Jun 2012 P3 / Dec 2008 — Effective Access Time

- EAT = Hit x t_hit + Miss x t_miss.
- NET Dec 2008: t_assoc = 20 ns, page table = 100 ns, EAT = 28 ns, h = ?
  → 28 = h(20) + (1-h)(100) to 28 = 100 - 80h to h = 72/80 = 0.9 = 90%.
- NET Jun 2012 (i, j, k): हर k instruction पर extra j - औसत extra = j/k - i + j/k.
- NET Jun 2013 Q: EAT = 20 ns + (10 ms/10^6) = 20 + (10^7 ns/10^6) = 20 + 10 = 30 ns.

(2) NET Jun 2012 — FIFO page faults

- Trace: 4,3,2,1,4,3,5,4,3,2,1,5, frames = 4.
  → 4 F(4) | 3 F(4,3) | 2 F | 1 F to 4 faults (frames 4,3,2,1)
  → 4 hit | 3 hit | 5 F (4 निकला: 3,2,1,5) | 4 F (3 निकला: 2,1,5,4) | 3 F (2 निकला: 1,5,4,3) | 2 F (1 निकला: 5,4,3,2) | 1 F (5 निकला: 4,3,2,1) | 5 F (4 निकला: 3,2,1,5).
- Total faults = 4 + 6 = 10.

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याद रखने की ट्रिक

★ OS types: B-M-T-D-R-E = Batch, Multiprog, Time-share, Distributed, Real-time, Embedded

★ Oracle OS नहीं, DBMS है: 'Oracle का काम data, OS का काम dada-giri' (hardware boss)

★ nice = priority set, kill = signal; Net Shell UNIX में नहीं होता

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चित्र से समझो: Multiprogramming vs Time-sharing

Multiprogramming

- कई programs memory में
- I/O wait पर CPU दूसरे को
- Goal: CPU utilization

VS

Time-sharing

- छोटे time slice (Round Robin)
- हर user को तुरंत response
- Goal: कम response time

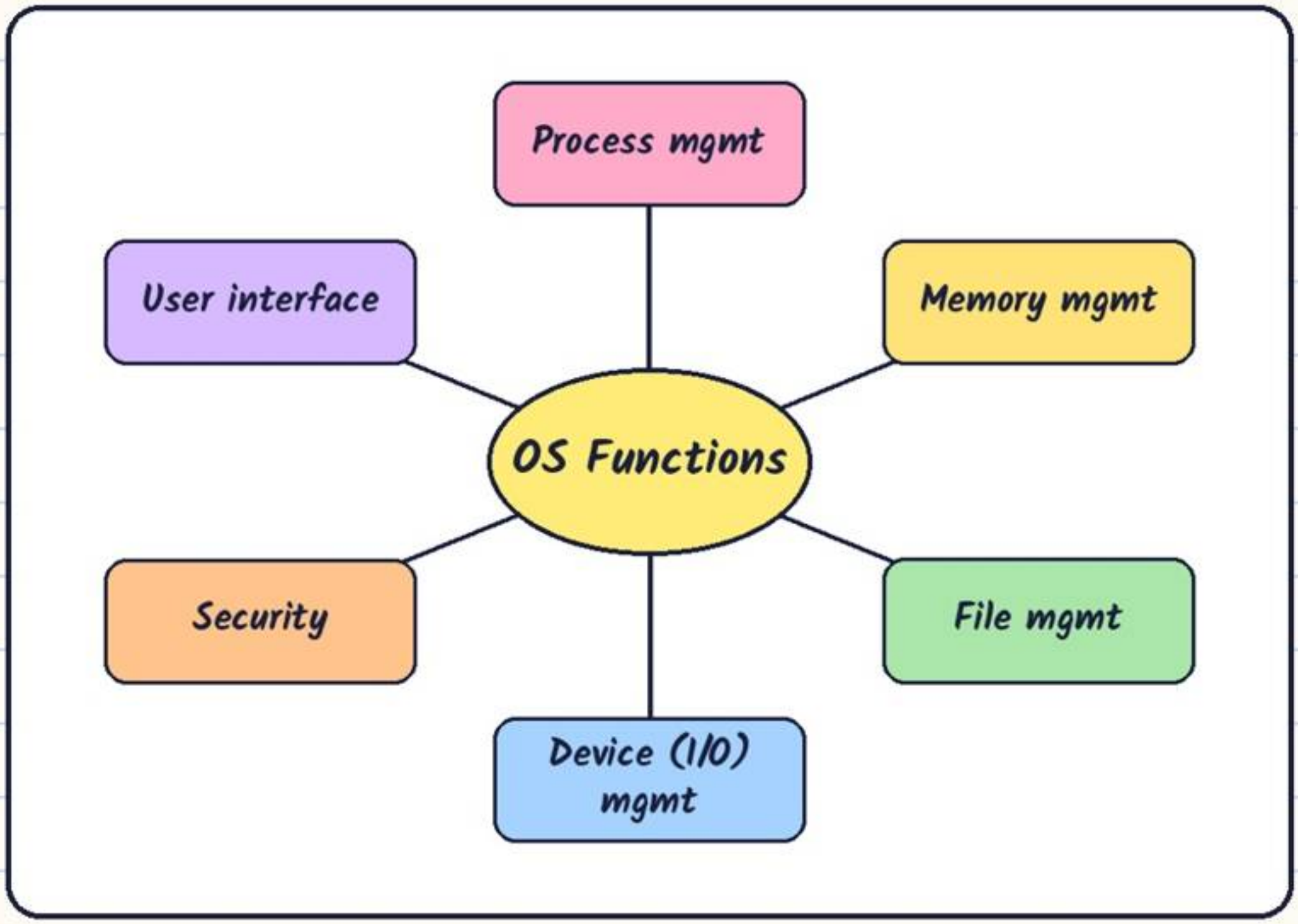
★ Multiprogramming = CPU utilization बढ़ाओ

★ Time-sharing = छोटे time slice, कम response time

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चित्र से समझो: OS के कार्य



★ OS के 6 मुख्य functions याद रखो

★ Process, Memory, File, I/O, Security, UI

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अब Test करो: 25 सवाल

1. Technique that marks the beginning of computer communications? (TRE 1.0)

(A) User Environment
(B) Batch Environment
(C) Time Sharing
(D) More than one
(E) None of these

2. _____ is used to shift processes so they are contiguous and all free memory is in one block. (TRE 1.0)

(A) Fragmentation
(B) Compaction
(C) External Fragment.
(D) More than one
(E) None of these

3. Which of the following is not an operating system? (TRE 2.0)

(A) Linux
(B) Windows
(C) Oracle
(D) More than one
(E) None of these

4. Software that runs on hardware and serves as a platform for other software is called (TRE 2.0)

(A) System software
(B) Operating system
(C) Application software
(D) More than one
(E) None of these

5. Key(s) used to delete a file without sending it to Recycle Bin (TRE 2.0)

(A) Delete
(B) Shift + Delete
(C) Ctrl + Delete
(D) More than one
(E) None of these

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Test जारी: सवाल 6-10

6. Everything below the system call interface and above the physical hardware is known as _____. (NET Dec 2012)

(A) Kernel
(B) Bus
(C) Shell
(D) Stub

7. In UNIX, which of the following command is used to set the task priority ? (NET Dec 2012)

(A) init
(B) nice
(C) kill
(D) PS

8. Which of the following is a type of operating system? (TRE 3.0)

(A) Real-time OS
(B) Embedded OS
(C) Network OS
(D) More than one
(E) None of these

9. Which is NOT a memory management technique used by OS? (TRE 3.0)

(A) Paging
(B) Segmentation
(C) Fragmentation
(D) More than one
(E) None of these

10. Which is a file system commonly used in operating systems? (TRE 3.0)

(A) NTFS
(B) FAT32
(C) HFS+
(D) More than one
(E) None of these

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Test जारी: सवाल 11-15

11. Which one of the following options is not a shell in UNIX system? (NET Jun 2012)

(A) Bourne Shell
(B) C Shell
(C) Net Shell
(D) Korn Shell

12. Which of the following memory allocation scheme suffers from external fragmentation ? (NET Dec 2012)

(A) Segmentation
(B) Pure demand paging
(C) Swapping
(D) Paging

13. What is a page in paging? (TRE 3.0)

(A) Fixed block of virtual mem
(B) Fixed block of physical mem
(C) A memory allocation type
(D) More than one
(E) None of these

14. Page size 4 KB, physical address space 64 GB; bits needed for page offset? (TRE 3.0)

(A) 10 bits
(B) 12 bits
(C) 14 bits
(D) More than one
(E) None of these

15. What is a segment base in segmentation? (TRE 3.0)

(A) Start addr in physical mem
(B) Start addr in virtual mem
(C) End addr in physical mem
(D) More than one
(E) None of these

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Test जारी: सवाल 16-20

16. What causes thrashing to occur? (TRE 3.0)

(A) Excessive paging
(B) Insufficient disk space
(C) Hardware failures
(D) More than one
(E) None of these

17. Assume N segments in memory and a page size of P bytes. The wastage on account of internal fragmentation is : (NET Jun 2009)

(A) NP/2 bytes
(B) P/2 Bytes
(C) N/2 Bytes
(D) NP Bytes

18. Semaphores S,Q=1. P1: wait(S),wait(Q); P2: wait(Q),wait(S). यह स्थिति कहलाती हैं? (NET Jun 2013)

(A) Starvation
(B) Race condition
(C) Multithreading
(D) Deadlock

19. Physical memory 8 GB, virtual 16 GB. Using 6 GB physical and 10 GB virtual: memory swapped out? (TRE 3.0)

(A) 2 GB
(B) 4 GB
(C) 6 GB
(D) More than one
(E) None of these

20. How is the TLB typically organized? (TRE 3.0)

(A) Direct-mapped cache
(B) Associative cache
(C) Set-associative cache
(D) More than one
(E) None of these

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Test जारी: सवाल 21-25

21. Which of the following is a common file operation? (TRE 3.0)

(A) Create
(B) Delete
(C) Rename
(D) More than one
(E) None of these

22. Suppose it takes 100 ns to access page table and 20 ns to access associative memory. If the average access time is 28 ns, the corresponding hit rate is : (NET Dec 2008)

(A) 100 percent
(B) 90 percent
(C) 80 percent
(D) 70 percent

23. Page trace 4,3,2,1,4,3,5,4,3,2,1,5; FIFO, 4 frames: page faults = ? (NET Jun 2012)

(A) 8
(B) 9
(C) 10
(D) 12

24. Disk queue 98,183,37,122,14,124,65,67; head at 53; FCFS: total head movement? (TRE 3.0)

(A) 640
(B) 620
(C) 630
(D) More than one
(E) None of these

25. A thread shares with other threads of the same process its (TRE 3.0)

(A) Thread ID
(B) Program counter
(C) Code & data section
(D) More than one
(E) None of these

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